

"Individual" (i.e. "group") projects

1. Butterfly coloration evolution
2. Sheep movements
3. Driftwood self-regulation
4. Evacuation
5. Flu virus
6. Pasture and forest



1- Butterfly coloration evolution



How do colorations of butterflies emerge from predator-prey interactions and camouflage?

The colorful canvas of butterfly wings are examples of evolutionary innovation and adaptation. Starting in the 19th century, naturalists investigated the function of wing-color patterns in different butterfly species. The understanding of population genetics in the early 20th century led to associations between specific genes and color patterns, helping elucidate how and when they evolved. Recently, researchers have discovered that just a few genes are responsible for setting up the patterning throughout the wing.

The environmental conditions (dominant color in the surroundings) and predator avoidance through camouflage are key selective drivers of colorations. Observations of black and white colorations of closely related butterflies living in different habitats are examples of local adaptations and natural selection. For instance, peppered moths are displaying three morphs: a white morph, a black morph, and different intermediate shades of gray.

A research question is to evaluate how predatory selection in a gradient of environmental conditions (black to white) can allow the emergence of distinct populations with different colors. Create an agent-based model describing the butterfly population, its reproduction at a fixed rate, with coloration dependent on a simple transmission system (each parents transmit black or white depending on its color: white butterflies only transmit white, black only black and gray ones have 50% of chance to transmit either color), predation pressure (modeling a predator population that feeds on butterflies but can be tricked by their camouflage) and an abrupt or gradual transition in local environmental colors to explore under which conditions distinct morphs can emerge.

Extension 1: Modify the model by considering that predators prefer to hunt the most represented color of the moth population.

Extension 2: Modify the model to include dynamic environmental transitions, where the colors of the environmental patches can change at a given speed. Investigate how the variability in this change rate influences the emergence and stability of distinct morphs in butterfly populations.

2- Sheep movements



How do herd behavior and pathway tracks emerge from individual sheep behaviors?

Sheep typically display herding behaviors by following and keeping in close distance from each other. These behaviors may be an evolutionary response to predator risks and resource use optimization. The herd form and collective movement dynamics are emerging from the individual behaviors. Another emergent characteristic is that the herds follow pathways when moving in heterogeneous landscape with obstacles where they cannot go.

A research question is to assess the effects of obstacle spatial distribution on the emergence of frequently used pathways. An agent-based model describing the sheeps, their interactions, the obstacles and the pathway taken by the sheep (number of passages in an area or tramping) may serve to explore how obstacle distribution influences the emergence of pathway tracks in the landscape.

Extension 1: Extend the agent-based model to incorporate variations in the obstacle types, such as different sizes, shapes, or levels of complexity, to assess their distinct impacts on pathway emergence and utilization by the herding sheep.

Extension 2: Introduce environmental factors, such as specific vegetation attracting some sheeps or topography making some path less easy to use, into the model to examine how these factors interact with obstacle spatial distribution in shaping the emergence of pathways in the landscape.

Extension 3: Introduce a new species called *dog* which will scare sheeps and examine how these factors interact with spatial distribution.

3- Driftwood regulation

Self-regulating access to natural resources

On the coast of a distant country, people compete for the gathering of driftwood brought to the shore by storms. Whoever is first onto a stretch of the shore after high tide is allowed to take whatever he wishes up to its carrying capacity and to gather it into a pile above the high-tide line. To indicate ownership, piles are marked by placing two stones on their top. The wood it contains is then regarded as the property of a driftwood collector. Only wood pile owners always respect pile ownership. Collectors having not yet established a pile can head towards wood piles and “steal” the wood it contains, but only when no pile owner is observing them.

An ABM based on this description will be used to explore the value of this “peer-pressure” regulation in addressing wood theft. Is it possible, without any external enforcement, to reach a stabilized situation?

Extension 1: Modify the agent-based model to explore variations in the recognition of pile ownership. Introduce the ability to steal to owners and analyze the impact on the stability of pile ownership over time.

Extension 2: Extend the model to introduce external factors, such as the arrival of external authorities or external enforcement mechanisms. Explore how the introduction of external influences affects the stability of the system and the behaviors of wood collectors.

Extension 3: Conduct an exploration exercise by varying the size of the wood collector groups. Investigate how the size of collector groups influences the emergence of stabilized situations, considering aspects like cooperation, competition, and the prevention of wood theft. Analyze the system's resilience to perturbations based on group size.



4- Evacuation



How to better manage the pedestrian evacuation of a population on a beach in a tsunami context?

Populations are increasingly vulnerable to disastrous natural or technological events, as demographic and urban growth lead to greater exposures of goods and people. Hanoi, for example, is particularly hard hit by flooding. Some districts on the banks of the Red River are also threatened by potential dike breaching. In the event of a levee failure, it is important to be able to evacuate the population living in these areas before the water arrives.

The goal of this project is to build an agent-based model of people evacuation. In this model, flooding will not be modeled by itself, just the behavior of residents in the face of the threat. People will only evacuate if they have been informed of the imminent risk of flooding. At the start of the simulation, we assume that all residents are located in their own homes and only 10% of the population (randomly chosen) will be aware of this information. Once informed, people will evacuate to the shelter (the largest building in the area). A person observing someone evacuating (at a distance of less than 10m) will have a probability of 0.1 of evacuating in turn. The simulation finished when all people had evacuated (the ones who got the information).

For building and road data, you can use the shapefiles used in previous exercises, or download new data as discussed in class.

Extension 1: not all residents know where to evacuate and only 10% will go directly to the shelter. The rest will move to random buildings trying to search for the shelter - if the resident is less than 20 m from the shelter, he/she goes straight to it.

Extension 2: We consider different types of mobility: car, motorcycle, walking. A person is 20% likely to have a car, 70% likely to have a motorcycle. A car can go faster on an open road, but will be more affected by traffic jams. Motorcycles will travel slightly slower than cars (0.85 factor), but will be half as affected by traffic jams. Pedestrians will be 10 times slower than cars, but will be 5 times less affected by traffic jams.

5- Flu Virus



How can local isolation reduce the diffusion of an epidemics?

Influenza, commonly known as "flu", is an infectious disease caused by *influenza viruses*. The symptoms begin from one to four days after exposure to the virus (typically two days) and last for about 2–8 days. Here we consider that the disease is not fatal: after a symptomatic period, the patient recovers.

In flu epidemics, healthy people (also called *Susceptibles*) may become sick (*Infected*) when they meet other people already carrying the disease. In turn, they contribute to spreading the disease when they meet other people. The probability to become infected when in contact with an infected person is 33%. The recovery rate can be a fixed infectious period of 10 days or a probability of 10% to recover each day. After this period, a sick person recovers.

The aim of this project is to propose a model to represent the spread of a disease in a city (using GIS data of buildings and roads). In the morning, inhabitants go to work: they move from their home to their workplace (chosen randomly among the buildings) before returning home in the evening. We consider that local authorities can test people (1% of the population every day). When a person tested is infected, he or she is isolated at home for 12 days.

For building and road data, you can use the shapefiles used in previous exercises, or download new data as discussed in class.

Extension 1: add the notion of family. Each building houses a family of 3 to 6 people (0-2 children). The city's largest building is a school, where children spend their day.

Extension 2: Add the notion of vaccination: every day, the authorities can vaccinate 0.05% of the population (unvaccinated people only). Once vaccinated, a person's probability of contagion is divided by 3.

6- Pasture and Forest



Shepherds graze their herds (a species of goat called *Logo*) in the undergrowth of a forest with *Cormasis gama* as the dominant tree species. Patches of *Cormasis gama* form groves that are disseminated in a pasture. The feeding potential for *Logo* goats is higher in the forest than in the pasture, but *Cormasis gama* do not resist grazing by *Logo* goats: the cover then turns from forest to pasture. On the other hand, *Cormasis gama* trees recolonize fringe-pasture at the edge of a grove with a monthly probability ranging from 0.0025 to 0.02 proportionally to the relative abundance of *Cormasis gama* in the surrounding of the fringe-pasture area.

The grazing is controlled by the shepherds, based on the memorization of the places they visited in the past and also based on their attitude (respectful or disrespectful) toward the Forest Department in charge of protecting *Cormasis gama* trees. The grazing season lasts 10 months. Each month during the grazing season, shepherds must place their animals somewhere. In case there is no unoccupied place within their perception range, they remain at the same place, otherwise they move, preferably into the forest (more food for the *Logo* goats). Yet sometimes some of them will refrain from entering the forest, when the size of the *Cormasis gama* patch is too small according to their opinion.

Shepherds record the number of months they were able to perceive the forest during a grazing season. By the end of the grazing season, each shepherd computes a specific indicator, called “self-authorized minimum size”, set as the difference between the grazing season duration in months (10) and the number of months they were able to perceive the forest. When the next grazing season starts, these individual memories are reset. Relating only to their individual representation, shepherds will refrain from grazing within *Cormasis gama* groves whose size is lower than their current self-authorized minimum size.

By the end of each grazing season, the Forest Department determines its own authorized minimum size by requesting all the shepherds to report their self-authorized minimum size and computing the average value, which will determine the institutional-authorized minimum size. If the size of a *Cormasis gama* grove is below the institutional authorized minimum size, it will be marked “protected” for the next grazing season.

Respectful shepherds will indefectibly follow the regulation set by the Forest Department, never entering protected *Cormasis gama* groves. Disrespectful shepherds will only refer to their own perception to decide whether to graze a forest area, not caring about the regulation set by the Forest Department.

What would be the future of such a socio-ecosystem:

if all shepherds were to enter the forest without any regulation?

if all shepherds were to follow their idea about the minimum size?

if all shepherds were to follow the institutional minimum size?